

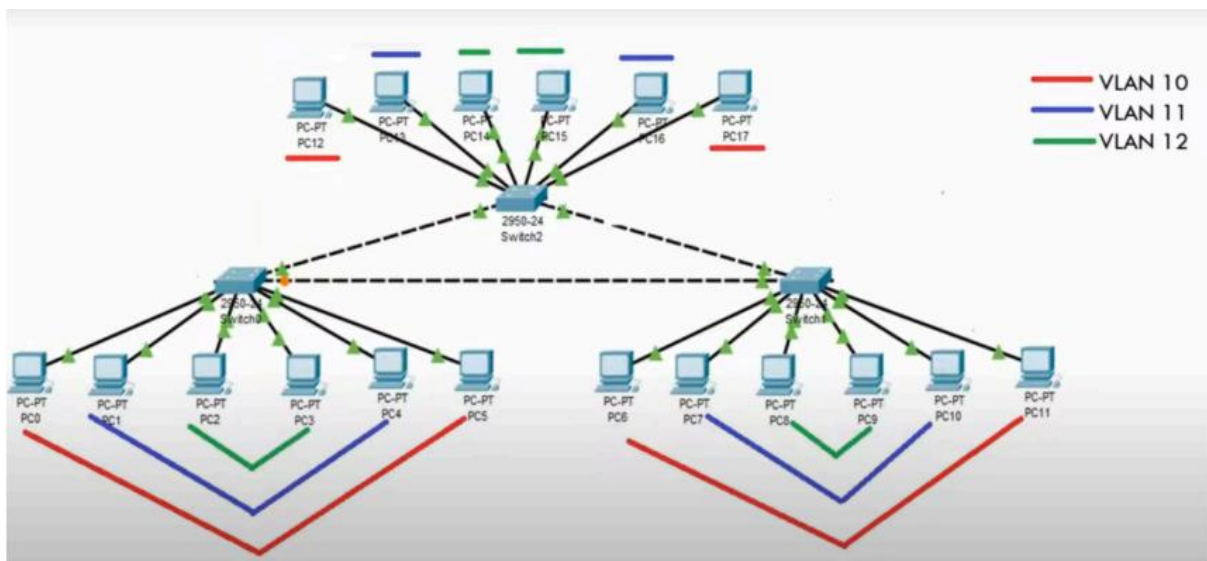
Computer Networks: Assignment-6

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Q 1) Construct the below network in the Cisco packet tracer, as shown below, and connect them more securely using VLAN.

Diagram:



IP Config:

PC0: 192.168.10.1 / 255.255.255.0

PC1: 192.168.20.1 / 255.255.255.0

PC2: 192.168.30.1 / 255.255.255.0

PC3: 192.168.30.2 / 255.255.255.0

PC4: 192.168.20.2 / 255.255.255.0

PC5: 192.168.10.2 / 255.255.255.0

PC6: 192.168.10.3 / 255.255.255.0

PC7: 192.168.20.3 / 255.255.255.0

PC8: 192.168.30.3 / 255.255.255.0

PC9: 192.168.30.4 / 255.255.255.0

PC10: 192.168.20.4 / 255.255.255.0

PC11: 192.168.10.4 / 255.255.255.0

PC12: 192.168.10.5 / 255.255.255.0

PC13: 192.168.20.5 / 255.255.255.0

PC14: 192.168.30.5 / 255.255.255.0

PC15: 192.168.30.6 / 255.255.255.0

PC16: 192.168.20.6 / 255.255.255.0

PC17: 192.168.10.6 / 255.255.255.0

Switch Config:

The image displays two screenshots of a network switch configuration interface for Switch1. The top screenshot shows the configuration for FastEthernet0/1, and the bottom screenshot shows the configuration for FastEthernet0/2. Both interfaces are configured as Access ports in VLAN 10 and 20 respectively, with Port Status, Bandwidth, and Duplex settings.

FastEthernet0/1 Configuration:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- Access:
- VLAN:
- Tx Ring Limit:

FastEthernet0/2 Configuration:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- Access:
- VLAN:
- Tx Ring Limit:

Switch1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/3

Port Status

Bandwidth

Duplex

Access

Tx Ring Limit

100 Mbps

10 Mbps

Half Duplex

Full Duplex

VLAN

30

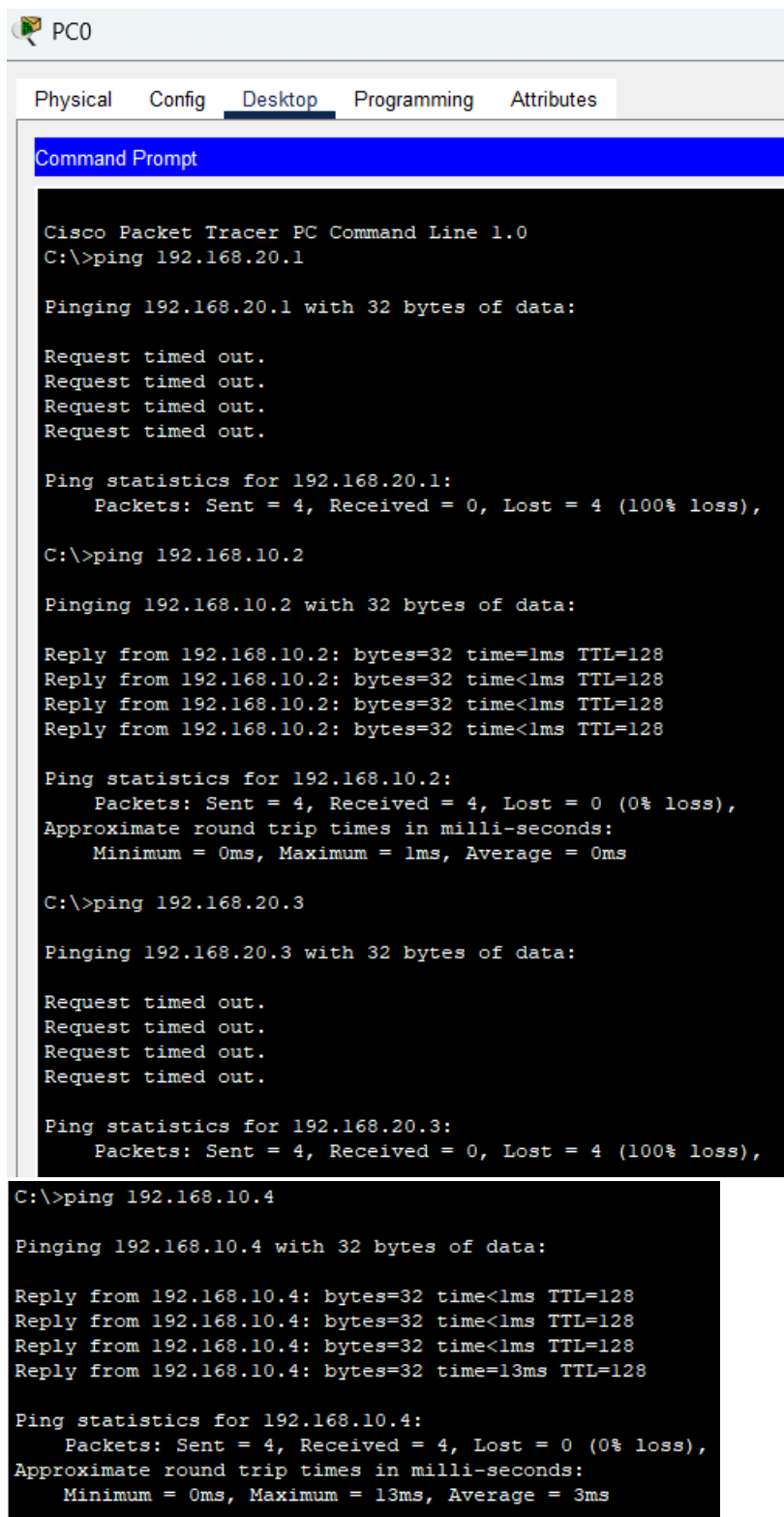
10

On

Auto

Auto

Output:



```
PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Q 2) Design s network in CISCO packer tracer to connect Accounts, Sales, Delivery, and Manufacturing departments:

- a) Each department should contain at least 2 PCs.
- b) An appropriate number of switches and routers should be used in the network, and proper IP addresses with subnet masks should be given.
- c) Connect all the devices with the appropriate wires and check their connectivity.
- d) Check the OSI layer information of the packet as shown below:

PDU Information at Device: Switch1

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Switch1

Source: PC0

Destination: PC3

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Ethernet II Header 000C.CF0B.4701 >> 000C.859C.AD49

Layer 1: Port FastEthernet0/3

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Ethernet II Header 000C.CF0B.4701 >> 000C.859C.AD49

Layer 1: Port(s): FastEthernet0/2

1. The frame source MAC address was found in the MAC table of Switch.

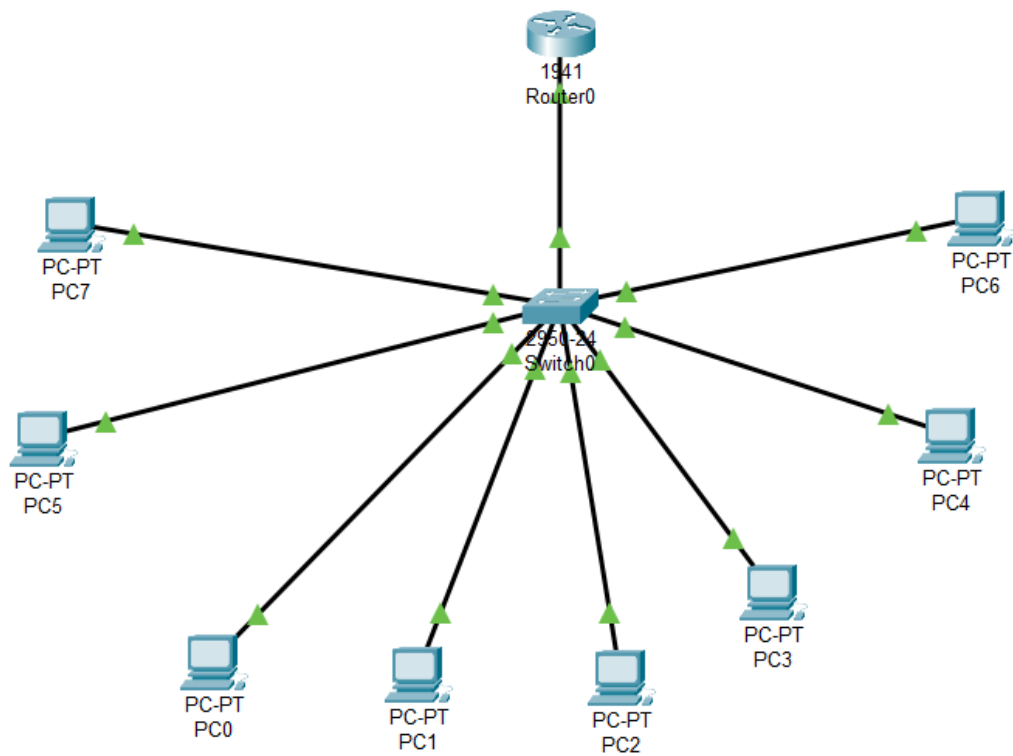
2. This is a unicast frame. Switch looks in its MAC table for the destination MAC address.

Challenge Me

<< Previous Layer

Next Layer >>

Diagram:



IP Config:

PC0: 192.168.10.10 / 255.255.255.0

PC1: 192.168.10.11 / 255.255.255.0

PC2: 192.168.20.10 / 255.255.255.0

PC3: 192.168.20.11 / 255.255.255.0

PC4: 192.168.30.10 / 255.255.255.0

PC5: 192.168.30.11 / 255.255.255.0

PC6: 192.168.40.10 / 255.255.255.0

PC7: 192.168.40.11 / 255.255.255.0

Switch Config:

The image displays three sequential screenshots of a network switch configuration window titled "Switch0". Each screenshot shows the "Config" tab selected, with a left-hand navigation pane and a main configuration area for a specific FastEthernet interface.

Navigation Pane (Common to all):

- GLOBAL
 - Settings
 - Algorithm Settings
- SWITCHING
 - VLAN Database
- INTERFACE
 - FastEthernet0/1
 - FastEthernet0/2
 - FastEthernet0/3
 - FastEthernet0/4
 - FastEthernet0/5
 - FastEthernet0/6

FastEthernet0/2 Configuration:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- Access:
- VLAN:
- Tx Ring Limit:

FastEthernet0/4 Configuration:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- Access:
- VLAN:
- Tx Ring Limit:

FastEthernet0/6 Configuration:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
- Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto
- Access:
- VLAN:
- Tx Ring Limit:

Output:

PDU Information at Device: Switch0



OSI Model

Outbound PDU Details

At Device: Switch0
Source: Switch0
Destination: STP Multicast Address

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3

Layer2

Layer1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: IEEE 802.3 Header 0060.2F4D.
2C01 >> 0100.0CCC.CCCD LLC SNAP STP
BPDU
Layer 1: Port(s): FastEthernet0/1

1. The STP process sends out a configuration BPDU.
2. The device encapsulates the PDU into an Ethernet frame.
3. The Switch uses the first VLAN number used in the trunk as the outgoing VLAN number in the IEEE 802.1q Header.

Challenge Me

<< Previous Layer

Next Layer >>